Forced Mechanics of the Mixed-Boundary Torsion Problem:
Exact Laws, Sharp Star-Shaped Bounds, and the Closure of the Distance Conjecture

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Abstract

This paper records, with complete proofs, the forced mechanics of the mixed-boundary torsion problem $-\nabla\cdot(\mathcal{A}\nabla u) = q$ with $u = 0$ on a supply portion $\Sigma \subset \partial\Omega$ and the natural (conormal Neumann) condition on the remainder Γ; coefficients are measurable, symmetric, and uniformly elliptic, $P_{\min} I \leq \mathcal{A} \leq P_{\max} I$ a.e., with every constant below depending on $P_{\max}$ only. The results: (1) the exact one-dimensional law $u(x) = q\int_0^x (L-s)/A(s)\,ds$, with the sharp far-end bound $qL^2/(2P_{\max})$ and equality exactly for $A \equiv P_{\max}$; (2) the ball-comparison mean bound: for every ball $B_r \subseteq \Omega$, $\|u\|_{L^\infty} \geq q\,r^2/\big(n(n+2)P_{\max}\big)$ — constant 15 in dimension three — via same-operator comparison and torsional-rigidity anti-monotonicity; (3) the pointwise full-boundary bound $u(x) \geq q\,\mathrm{dist}(x,\partial\Omega)^2/(2nP_{\max})$ for constant tensors; (4) the sharp star-shaped theorem: if Γ is star-shaped about x_0, then $u(x_0) \geq (q/2n)\inf_\Sigma (z-x_0)^\top \mathcal{A}^{-1}(z-x_0) \geq q\,d_{\mathrm{E}}(x_0,\Sigma)^2/(2nP_{\max})$, ratio-free, with equality on balls, central reflection sectors, and ellipsoids; (5) the closure of the distance conjecture P2: an exactly solvable family of tapered horns yields $P_{\max}\|u\|_{L^\infty}/(qR^2) \leq (1-\delta^2+k\varepsilon^2/2)/\big((2+k(n-1))(1-\delta)^2\big)$, so all three metric variants of the conjectured bound $qR^2/(2nP_{\max})$, $R = \sup_x d(x,\Sigma)$, are false, the infimum of the coefficient over admissible data is 0, and an explicit Lipschitz instance attains the exact rational value bound $423/2890 < 1/6$ in dimension three; and (6) the dichotomy within the horn family: the star condition degenerates exactly at the straight cone $k = 2$, where the coefficient equals $1/(2n)$ in every dimension — a sharp *sufficient* hypothesis whose removal beyond the cone permits arbitrarily severe failure (necessity over all geometries is not asserted). Distance to the full boundary always controls u from below; distance to the supply portion does so under the star condition; on the horns beyond it, the governing quantity is the volume-graded resistance, for which they provide the exact identity. Only forced material appears; every displayed identity and rational inequality has additionally been machine-verified in exact arithmetic.

1 Setting and conventions

Standing conventions. Throughout, $n \geq 2$ unless stated otherwise, $q > 0$ is constant, and $\mathcal{A}$ denotes a measurable symmetric coefficient tensor field which is uniformly elliptic: $P_{\min} I \leq \mathcal{A} \leq P_{\max} I$ a.e. for fixed constants $0 < P_{\min} \leq P_{\max}$. The lower bound enters only through well-posedness (coercivity); no constant in any estimate below depends on it. We freely use classical facts — Lax–Milgram, Rellich compactness, trace theory on Lipschitz boundaries, Stampacchia's truncation identities, and interior elliptic regularity for constant-coefficient operators — and prove everything else. "Ratio-free" means that a constant depends on $P_{\max}$ only, never on the ellipticity ratio $P_{\max}/P_{\min}$, the volume, or regularity moduli.

Definition 1.1 (Mixed torsion problem, trace formulation)**.** *Let $\Omega \subset \mathbb{R}^n$ be a bounded connected Lipschitz domain and let $\partial\Omega = \overline{\Sigma} \cup \overline{\Gamma}$ with Σ relatively open of positive surface measure, and let $\mathcal{A}$ be measurable, symmetric, and uniformly elliptic ($P_{\min} I \leq \mathcal{A} \leq P_{\max} I$ a.e., $P_{\min} > 0$). Set*

$$V \;=\; \{\varphi \in H^1(\Omega) \;:\; \mathrm{tr}\,\varphi = 0 \;\sigma\text{-a.e. on } \Sigma\}.$$

The torsion function $u \in V$ *is the unique solution of*

$$B[u,\varphi] \;:=\; \int_\Omega \nabla\varphi\cdot\mathcal{A}\nabla u\,dz \;=\; \int_\Omega q\,\varphi\,dz \qquad \text{for all } \varphi \in V, \tag{1}$$

which exists by the standard Poincaré inequality on V, coercivity from uniform ellipticity, and Lax–Milgram. Formulation (1) *encodes $u = 0$ on Σ and the natural condition $\mathcal{A}\nabla u \cdot \nu = 0$ on Γ.*

Lemma 1.2 (Positivity and nontriviality)**.** *$u \geq 0$ a.e., and $u \not\equiv 0$.*

Proof. $u^- := \max(-u, 0) \in V$ (its trace vanishes on Σ). Then $B[u, u^-] = -\int_{\{u<0\}} \nabla u \cdot \mathcal{A}\nabla u \leq 0$ while $B[u, u^-] = \int q\, u^- \geq 0$; hence both vanish, $\nabla u^- \equiv 0$, and the constant u^- has zero trace on Σ, so $u^- \equiv 0$. If $u \equiv 0$ then (1) forces $\int q\varphi = 0$ for all $\varphi \in V$, impossible. □

Metrics and the distance conjecture. Write $d_{\mathrm{E}}(z, \Sigma) = \inf_{w\in\Sigma} |z - w|$ for the Euclidean distance, $d_\Omega(z, \Sigma)$ for the infimal length of rectifiable paths inside Ω from z to Σ, and, for locally Lipschitz f on Ω,

$$d_{\mathcal{A}}(z, \Sigma) \;=\; \sup\Big\{ f(z) \;:\; \nabla f \cdot \mathcal{A}\nabla f \leq 1 \text{ a.e.},\;\; \limsup_{w\to\Sigma} f(w) \leq 0 \Big\} \tag{2}$$

for the operator-intrinsic distance. With $R_\bullet = \sup_{z\in\Omega} d_\bullet(z, \Sigma)$, the *distance conjecture* P2 asserted, for all admissible data,

$$\text{(P2-E)}\;\; \|u\|_{L^\infty} \;\geq\; \frac{q\, R_{\mathrm{E}}^2}{2n\, P_{\max}}, \qquad \text{(P2-G)}\;\; \|u\|_{L^\infty} \;\geq\; \frac{q\, R_\Omega^2}{2n\, P_{\max}}, \qquad \text{(P2-A)}\;\; \|u\|_{L^\infty} \;\geq\; \frac{q\, R_{\mathcal{A}}^2}{2n}. \tag{3}$$

Since $d_{\mathrm{E}} \leq d_\Omega \leq \sqrt{P_{\max}}\, d_{\mathcal{A}}$, the variants are nested: P2-A ⇒ P2-G ⇒ P2-E. Sections 2–4 establish what *does* hold; Sections 5–6 close (3) and exhibit, within an explicit family, the exact transition between the star-controlled regime and unbounded failure.

Lemma 1.3 (Metric collapse for constant scalar coefficients)**.** *If $\mathcal{A} = P\,I$ with constant $P > 0$, then $d_{\mathcal{A}}(z, \Sigma) = d_\Omega(z, \Sigma)/\sqrt{P}$ for every $z \in \Omega$.*

Proof. "$\geq$": $f_0 := d_\Omega(\cdot, \Sigma)/\sqrt{P}$ is admissible in (2): small balls in Ω are convex, so d_Ω is locally 1-Lipschitz, hence $|\nabla f_0| \leq P^{-1/2}$ a.e., i.e. $\nabla f_0 \cdot \mathcal{A}\nabla f_0 \leq 1$; and $f_0 \to 0$ at Σ. "$\leq$": any admissible f has $|\nabla f| \leq P^{-1/2}$ a.e., hence is $P^{-1/2}$-Lipschitz along every rectifiable path in Ω; integrating along paths from z that approach Σ and using $\limsup_\Sigma f \leq 0$ gives $f(z) \leq d_\Omega(z, \Sigma)/\sqrt{P}$. □

2 The one-dimensional law

Theorem 2.1 (Exact 1D law)**.** *Let $0 < P_{\min} \leq A(x) \leq P_{\max}$ be measurable on $(0, L)$ and $q > 0$. The problem*

$$-\big(Au'\big)' = q \;\text{ on } (0, L), \qquad u(0) = 0, \qquad (Au')(L) = 0,$$

has the unique solution (with Au' absolutely continuous)

$$u(x) \;=\; q \int_0^x \frac{L - s}{A(s)}\, ds.$$

Consequently u is nondecreasing,

$$u(x) \;\geq\; \frac{q\, x^2}{2P_{\max}} \quad \text{for every } x \in [0, L], \qquad \|u\|_{L^\infty} \;=\; u(L) \;=\; q \int_0^L \frac{L - s}{A(s)}\, ds \;\geq\; \frac{qL^2}{2P_{\max}},$$

and equality holds in the far-end bound if and only if $A = P_{\max}$ a.e. on $(0, L)$.

Proof. Existence. With u as displayed, $A(x)u'(x) = q(L - x)$ by construction; this is absolutely continuous, vanishes at $x = L$, and $-(Au')' = q$ a.e.; $u(0) = 0$. *Uniqueness.* A difference v of two solutions satisfies $(Av')' = 0$, so $Av' \equiv c$; the far-end condition gives $c = 0$, hence $v' = 0$ a.e. and $v \equiv v(0) = 0$. *Monotonicity and bounds.* $u' = q(L - x)/A \geq 0$. Since $A \leq P_{\max}$ and $L - s \geq x - s$ on $[0, x]$, $u(x) \geq \frac{q}{P_{\max}} \int_0^x (x - s)\, ds = qx^2/(2P_{\max})$; at $x = L$, $u(L) \geq \frac{q}{P_{\max}} \int_0^L (L-s)\, ds = qL^2/(2P_{\max})$, with equality iff $\int_0^L (L-s)\big(\frac{1}{A} - \frac{1}{P_{\max}}\big) ds = 0$; the integrand is nonnegative and $L - s > 0$ a.e., forcing $A = P_{\max}$ a.e. □

Remark 2.2. Here $\Sigma = \{0\}$ and $d(x, \Sigma) = x$, so the pointwise bound reads $u \geq q\, d(\cdot, \Sigma)^2/(2P_{\max})$: the constant matches $2n$ at $n = 1$; the pointwise bound is strict for every $x < L$. The sink in Theorem 2.1 is q per unit *length*; the volume-graded 1D balance, with sink q per unit *volume* in a tube of varying cross-section, is a different law and is the mechanism of Section 5 (see Remark 7.1).

3 Ball comparison and the mean bound

Lemma 3.1 (Same-operator domain comparison). *Let $\omega \subseteq \Omega$ be open and let $w \in H_0^1(\omega)$ be the full-Dirichlet torsion function of ω for the* same *coefficient $\mathcal{A}$ and sink q: $\int_\omega \nabla\varphi \cdot \mathcal{A}\nabla w = \int_\omega q\varphi$ for all $\varphi \in H_0^1(\omega)$. Then $u \geq w$ a.e. on ω.*

Proof. Let $w_j \in C_c^\infty(\omega)$ with $w_j \to w$ in $H_0^1(\omega)$. Since $u \geq 0$ (Lemma 1.2), $(w_j - u)^+$ vanishes wherever $w_j = 0$, so it is an H^1 function supported in $\operatorname{supp} w_j \Subset \omega$; hence $(w_j - u)^+ \in H_0^1(\omega)$, and by truncation continuity $\varphi := (w-u)^+ \in H_0^1(\omega)$. Its extension by zero to Ω lies in V. Testing (1) with the extension and the ω-problem with φ and subtracting,

$$0 \;=\; \int_\omega \nabla\varphi \cdot \mathcal{A}\nabla(w-u) \;=\; \int_{\{w>u\}} \nabla(w-u)\cdot\mathcal{A}\nabla(w-u) \;\geq\; 0,$$

so $\nabla\varphi \equiv 0$ and $\varphi \in H_0^1(\omega)$ is constant, hence $\varphi \equiv 0$: $w \leq u$ a.e. on ω. $\square$

Lemma 3.2 (Torsional-rigidity anti-monotonicity). *Let ω be bounded open and let w, w' be the full-Dirichlet torsion functions of ω for coefficients $\mathcal{A} \leq \mathcal{A}'$ (a.e., as quadratic forms) and the same q. Then $\int_\omega w \geq \int_\omega w'$.*

Proof. For $\varphi \in H_0^1(\omega)$ set $J_{\mathcal{A}}(\varphi) = 2q\int\varphi - \int \nabla\varphi\cdot\mathcal{A}\nabla\varphi$. Since $B_{\mathcal{A}}[w,\varphi] = q\int\varphi$,

$$J_{\mathcal{A}}(\varphi) \;=\; 2B_{\mathcal{A}}[w,\varphi] - B_{\mathcal{A}}[\varphi,\varphi] \;=\; B_{\mathcal{A}}[w,w] - B_{\mathcal{A}}[\varphi - w, \varphi - w] \;\leq\; B_{\mathcal{A}}[w,w] \;=\; q\int_\omega w,$$

with equality at $\varphi = w$: thus $q\int w = \max J_{\mathcal{A}}$. Since $J_{\mathcal{A}}(\varphi) \geq J_{\mathcal{A}'}(\varphi)$ pointwise in φ, $q\int w = \max J_{\mathcal{A}} \geq \max J_{\mathcal{A}'} = q\int w'$. $\square$

Theorem 3.3 (Ball mean bound; the P1 constant $n(n+2)$). *Let u be as in Definition 1.1, with measurable, uniformly elliptic $\mathcal{A} \leq P_{\max} I$. For every open ball $B_r(x_0) \subseteq \Omega$,*

$$\frac{1}{|B_r|}\int_{B_r(x_0)} u \;\geq\; \frac{q\,r^2}{n(n+2)\,P_{\max}}, \qquad \textit{hence} \qquad \|u\|_{L^\infty} \;\geq\; \frac{q\,r^2}{n(n+2)\,P_{\max}}.$$

In particular $\|u\|_{L^\infty} \geq q\,r_{\mathrm{in}}(\Omega)^2/(n(n+2)P_{\max})$ with r_{in} the inradius, which is attained: $\operatorname{dist}(\cdot,\partial\Omega)$ *is continuous on the compact $\overline{\Omega}$, so the open ball of radius r_{in} about a maximizer lies in Ω. For $n = 3$ the constant is* 15.

Proof. Let $w_{\mathcal{A}}$ be the full-Dirichlet torsion of $B := B_r(x_0)$ for $\mathcal{A}$, and $w_{P_{\max}}$ that for $P_{\max}I$, namely

$$w_{P_{\max}}(z) \;=\; \frac{q\,(r^2 - |z-x_0|^2)}{2nP_{\max}}$$

(direct verification: $-P_{\max}\Delta w_{P_{\max}} = q$, zero boundary values). By Lemma 3.1, $u \geq w_{\mathcal{A}}$ on B; by Lemma 3.2 with $\mathcal{A}' = P_{\max}I$, $\int_B w_{\mathcal{A}} \geq \int_B w_{P_{\max}}$. The radial mean of $r^2 - |z-x_0|^2$ over B is

$$\frac{\int_0^r (r^2-\rho^2)\rho^{n-1}d\rho}{\int_0^r \rho^{n-1}d\rho} = \frac{r^{n+2}\left(\frac{1}{n} - \frac{1}{n+2}\right)}{r^n/n} = \frac{2r^2}{n+2},$$

so $\frac{1}{|B|}\int_B u \;\geq\; \frac{1}{|B|}\int_B w_{P_{\max}} \;=\; \frac{q}{2nP_{\max}}\cdot\frac{2r^2}{n+2} \;=\; \frac{qr^2}{n(n+2)P_{\max}}$, and the essential supremum dominates the mean. $\square$

Remark 3.4. Only $B \subseteq \Omega$ is used: the ball may touch $\partial\Omega$, including Σ, and the coefficient may be any measurable uniformly elliptic tensor. The bound is ratio-free and mean-form, not pointwise; the pointwise sharp bound for *constant* tensors is Corollary 4.4 below.

4 The star-shaped theorem

Throughout this section Ω is a bounded connected Lipschitz domain as in Definition 1.1, and $\mathcal{A}$ is a *constant* symmetric positive definite tensor with $\mathcal{A} \le P_{\max} I$; constancy makes uniform ellipticity automatic, with $P_{\min} = \lambda_{\min}(\mathcal{A})$.

Proposition 4.1 (Two identities)**.** *Fix $x_0 \in \Omega$ and set $f(z) = (z - x_0)^\top \mathcal{A}^{-1}(z - x_0)$. Then*

$$\mathcal{A}\nabla f = 2(z - x_0), \qquad \nabla\cdot(\mathcal{A}\nabla f) = 2n, \qquad \mathcal{A}\nabla f \cdot \nu = 2(z - x_0)\cdot \nu \quad \text{on } \partial\Omega,$$

and $f(z) \ge |z - x_0|^2/P_{\max}$, since in an orthonormal eigenbasis of $\mathcal{A}$ with eigenvalues $\lambda_i \le P_{\max}$, $f - |z - x_0|^2/P_{\max} = \sum_i w_i^2 (P_{\max} - \lambda_i)/(\lambda_i P_{\max}) \ge 0$.

Proof. $\nabla f = 2\mathcal{A}^{-1}(z - x_0)$, hence $\mathcal{A}\nabla f = 2(z - x_0)$, whose divergence is $2n$; the conormal identity and the eigenbasis decomposition are immediate. $\square$

Theorem 4.2 (Star-shaped lower bound)**.** *Let u be the torsion function of Definition 1.1 and let $x_0 \in \Omega$ satisfy the star condition*

$$(z - x_0)\cdot \nu(z) \;\ge\; 0 \qquad \text{for } \sigma\text{-a.e. } z \in \Gamma. \tag{4}$$

Then

$$u(x_0) \;\ge\; \frac{q}{2n} \inf_{z\in\Sigma} (z - x_0)^\top \mathcal{A}^{-1}(z - x_0) \;\ge\; \frac{q\, d_{\mathrm{E}}(x_0, \Sigma)^2}{2n\, P_{\max}}.$$

The constant $2n$ cannot be improved, and the bound is ratio-free: positive definiteness of $\mathcal{A}$ is used only qualitatively; the constant involves $P_{\max}$ alone. (Convex Ω satisfies (4) for every $x_0 \in \Omega$.)

Proof. Set $w = u + \frac{q}{2n} f$ with f as in Proposition 4.1. For $\varphi \in V$, Gauss–Green for the smooth field $\varphi\,\mathcal{A}\nabla f$ on the Lipschitz Ω gives

$$\int_\Omega \nabla\varphi \cdot \mathcal{A}\nabla f = -2n \int_\Omega \varphi + \int_\Gamma 2\,(z - x_0)\cdot\nu \;\operatorname{tr}\varphi\, d\sigma,$$

the Σ-part vanishing by the trace condition. Adding (1),

$$B[w, \varphi] \;=\; \frac{q}{n} \int_\Gamma (z - x_0)\cdot \nu \;\operatorname{tr}\varphi\, d\sigma \;\ge\; 0 \qquad \text{whenever } \varphi \in V,\ \varphi \ge 0,$$

by (4). Since f is continuous and $\overline{\Sigma}$ is compact,

$$\inf_{z\in\Sigma} f(z) \;=\; \min_{z\in\overline{\Sigma}} f(z).$$

Let $m = \frac{q}{2n}\inf_\Sigma f$ and $\varphi = (m - w)^+$. On Σ, $\operatorname{tr} w = \frac{q}{2n} f \ge m$, so $\operatorname{tr}\varphi = 0$ and $\varphi \in V$, $\varphi \ge 0$. Then

$$0 \;\le\; B[w, \varphi] \;=\; -\int_{\{w<m\}} \nabla w \cdot \mathcal{A}\nabla w \;\le\; 0,$$

so $\nabla\varphi \equiv 0$, and the constant φ has zero trace on Σ, forcing $\varphi \equiv 0$: $w \ge m$ a.e. Since $\mathcal{A}$ and q are constant, u is smooth in Ω; as $f(x_0) = 0$, $u(x_0) = w(x_0) \ge m$. The second inequality is $\inf_\Sigma f \ge \inf_\Sigma |z - x_0|^2/P_{\max} = d_{\mathrm{E}}(x_0, \Sigma)^2/P_{\max}$ by Proposition 4.1. Sharpness is Proposition 4.3. $\square$

Proposition 4.3 (Equality family)**.** *Equality holds in Theorem 4.2 in each of the following cases.*

(i) ***Ball.*** *$\Omega = B_R(x_0)$, $\Sigma = \partial B_R(x_0)$, $\mathcal{A} = P_{\max} I$: here $u = \dfrac{q\,(R^2 - |z - x_0|^2)}{2nP_{\max}}$ and $u(x_0) = \dfrac{qR^2}{2nP_{\max}}$.*

(ii) ***Central reflection sectors.*** *$\Omega = B_R(x_0) \cap C$ with C an open intersection of half-spaces whose boundary hyperplanes pass through x_0; Σ the spherical part, Γ the flat cuts, $\mathcal{A} = P_{\max} I$. The function of* (i)*, restricted, satisfies the equation, vanishes on Σ, and has conormal derivative $P_{\max}\nabla u \cdot \nu = -\frac{q}{n}(z - x_0)\cdot\nu = 0$ on each cut; by uniqueness it* is *the torsion function, and $u(x_0) = qR^2/(2nP_{\max})$ with* (4) *holding with equality on Γ.*

(iii) ***Ellipsoids (anisotropic sharpness).*** *$\Omega = \{f < r^2\}$, $\Sigma = \partial\Omega$, $\Gamma = \varnothing$: then $u = \frac{q}{2n}(r^2 - f)$ solves the problem and $u(x_0) = \frac{q}{2n}r^2 = \frac{q}{2n}\inf_\Sigma f$, attaining the first inequality of Theorem 4.2.*

Proof. Each candidate is smooth, lies in V, satisfies $-\nabla\cdot(\mathcal{A}\nabla u) = q$ by Proposition 4.1 (e.g. in (iii), $-\nabla\cdot(\mathcal{A}\nabla u) = \frac{q}{2n}\cdot 2n = q$), vanishes on Σ, and has zero conormal derivative on Γ where present; uniqueness of the weak solution identifies it with the torsion function. The stated values are direct evaluations. □

Corollary 4.4 (Distance to the full boundary always works). *Let u be as in Definition 1.1 with* constant *$\mathcal{A} \le P_{\max} I$, and let the Σ/Γ split be arbitrary. Then, for every $x_0 \in \Omega$,*

$$u(x_0) \;\ge\; \frac{q\,\mathrm{dist}(x_0, \partial\Omega)^2}{2n\,P_{\max}},$$

sharp on the ball with $\Sigma = \partial\Omega$.

Proof. Let $r = \mathrm{dist}(x_0, \partial\Omega)$ and $B = B_r(x_0) \subseteq \Omega$. By Lemma 3.1, $u \ge w_B$ a.e. on B, where w_B is the full-Dirichlet torsion of B for $\mathcal{A}$. Theorem 4.2 applied to $(B, \Sigma = \partial B, \Gamma = \varnothing)$ — the star condition is vacuous — gives $w_B(x_0) \ge \frac{q}{2n}\inf_{\partial B} f \ge qr^2/(2nP_{\max})$. Both u and w_B are continuous in B (constant coefficients), so the a.e. inequality holds at x_0. □

Remark 4.5 (Metric scope). Theorem 4.2 is *chordal*: it uses straight-line $\mathcal{A}$-distance to Σ, which dominates $d_{\mathrm{E}}^2/P_{\max}$. The star condition (4) does not force $d_\Omega(x_0,\Sigma) = d_{\mathrm{E}}(x_0,\Sigma)$, so no path-metric strengthening is asserted; likewise Corollary 4.4 is asserted for constant tensors only — for variable coefficients, only the mean bound of Theorem 3.3 is claimed. The contrast established by Sections 5–6: distance to $\partial\Omega$ bounds u unconditionally (Corollary 4.4); distance to Σ does so under the star condition — a sharp *sufficient* hypothesis whose removal permits arbitrarily severe failure.

5 Tapered horns: closure of the distance conjecture

Definition 5.1 (Truncated horn). *Fix $h > 0$, $k > 0$, $\varepsilon \in (0,1)$, $\delta \in (0,1)$, and set $g(x) = \varepsilon h\,(x/h)^{k/2}$ on $[0,h]$. Define*

$$\Omega_{k,\varepsilon,\delta} \;=\; \big\{(x,y) \in (\delta h, h) \times \mathbb{R}^{n-1} \;:\; |y| < g(x)\big\}, \qquad \Sigma \;=\; \{h\} \times \{|y| < \varepsilon h\},$$

with Γ the remaining boundary: the lateral wall $\{|y| = g(x)\}$ and the end face $F_\delta = \{\delta h\} \times \{|y| < g(\delta h)\}$. Take $\mathcal{A} = P\,I$ with constant $P > 0$. Since $g \in C^\infty([\delta h, h])$ has bounded slope and the wall meets the two flat faces transversally, $\Omega_{k,\varepsilon,\delta}$ is a bounded Lipschitz domain, so Definition 1.1 applies. Write $\rho = |y|$.

Proposition 5.2 (Exact solution with exactly vanishing wall flux). *Let*

$$a \;=\; \frac{q}{P\,\big(2 + k(n-1)\big)}, \qquad b \;=\; \frac{ak}{2}, \qquad v(x,y) \;=\; a\,(h^2 - x^2) \;-\; b\,|y|^2.$$

Then, identically:

(i) *$-P\Delta v = q$ on $\mathbb{R}^n$;*

(ii) *$\partial_\nu v = 0$ at every point of the lateral wall $\{\rho = g(x)\}$, $0 < x < h$;*

(iii) *on the end face F_δ (outward normal $-e_1$), $\partial_\nu v = -\partial_x v = 2a\delta h > 0$;*

(iv) *$\sup_{\overline{\Omega}_{k,\varepsilon,\delta}} v = v(\delta h, 0) = ah^2(1-\delta^2)$, and $v|_\Sigma = -b\rho^2 \in [-b(\varepsilon h)^2, 0]$.*

Proof. (i) $\Delta v = \partial_x^2 v + \Delta_y v = -2a - 2b(n-1)$, so $-P\Delta v = Pa\big(2 + k(n-1)\big) = q$. (ii) The wall is the zero set of $F(x,y) = |y| - g(x)$, with outward normal proportional to $(-g'(x),\, y/|y|)$. Since $\nabla v = (-2ax,\, -2by)$,

$$\nabla v \cdot (-g',\, y/|y|) \;=\; 2axg'(x) - 2b\rho \overset{\rho\equiv g}{=} 2axg'(x) - 2bg(x) \;=\; \big(ak - 2b\big)g(x) \;=\; 0,$$

using $xg'(x) = \frac{k}{2}g(x)$ for the power profile and $b = ak/2$. (iii)–(iv) are direct evaluations; v is strictly decreasing in x and in ρ. □

Theorem 5.3 (Comparison)**.** *On $\Omega_{k,\varepsilon,\delta}$, the torsion function satisfies*

$$0 \ \le\ u \ \le\ v + b(\varepsilon h)^2 \quad \text{a.e.}, \qquad \text{hence} \qquad \|u\|_{L^\infty} \ \le\ ah^2\big(1-\delta^2\big) + b(\varepsilon h)^2.$$

Proof. For $\varphi \in H^1(\Omega)$ the Gauss–Green identity on the Lipschitz domain, applied to the field $\varphi\, P\nabla v$ with v smooth, gives

$$B[v,\varphi] \ =\ \int_\Omega q\,\varphi \ +\ \int_{\partial\Omega} P\,(\partial_\nu v)\ \operatorname{tr}\varphi\, d\sigma.$$

By Proposition 5.2, the boundary integrand vanishes on the wall, equals $2a\delta h\, P \operatorname{tr}\varphi$ on F_δ, and for $\varphi \in V$ the Σ-part vanishes because $\operatorname{tr}\varphi = 0$ there. Hence, for $\varphi \in V$ with $\varphi \ge 0$ (whence $\operatorname{tr}\varphi \ge 0$),

$$B[u-v,\varphi] \ =\ -\,2a\delta h\, P \int_{F_\delta} \operatorname{tr}\varphi\, d\sigma \ \le\ 0.$$

Set $M = b(\varepsilon h)^2$ and $\varphi = (u-v-M)^+$. Its trace on Σ equals $(b\rho^2 - M)^+ = 0$ since $\rho \le \varepsilon h$ on Σ, so $\varphi \in V$, and $\varphi \ge 0$. By Stampacchia,

$$0 \ \ge\ B[u-v,\varphi] \ =\ \int_{\{u-v>M\}} \nabla(u-v)\cdot \mathcal{A}\nabla(u-v) \ \ge\ 0,$$

so $\nabla\varphi \equiv 0$; the constant φ has zero trace on Σ, hence $\varphi \equiv 0$ and $u \le v + M$. Combining with Lemma 1.2 and Proposition 5.2(iv) yields the L^∞ bound. □

Lemma 5.4 (Distances on the horn)**.** *For $z = (x,y) \in \Omega_{k,\varepsilon,\delta}$,*

$$d_{\mathrm{E}}(z,\Sigma) \ =\ d_\Omega(z,\Sigma) \ =\ h - x, \qquad \text{and} \qquad d_{\mathcal{A}}(z,\Sigma) = \frac{h-x}{\sqrt{P}}.$$

Consequently $R_{\mathrm{E}} = R_\Omega = \sqrt{P}\, R_{\mathcal{A}} = h(1-\delta)$.

Proof. The horizontal segment $t \mapsto (t,y)$, $t \in [x,h]$, stays in Ω because g is nondecreasing, and ends on $\overline{\Sigma}$; hence $d_\Omega \le h - x$. Any path from z to Σ has length at least its x-displacement $h - x$; and $d_{\mathrm{E}} \ge h - x$ for the same reason while $d_{\mathrm{E}} \le d_\Omega$. The last identity is Lemma 1.3. The supremum of $h - x$ over Ω is $h(1-\delta)$. □

Theorem 5.5 (Falsification on Lipschitz domains)**.** *On $\Omega_{k,\varepsilon,\delta}$,*

$$\frac{P_{\max}\,\|u\|_{L^\infty}}{q\,R^2} \ \le\ \beta(k,\varepsilon,\delta,n) \ :=\ \frac{1-\delta^2+\frac{k}{2}\varepsilon^2}{\big(2+k(n-1)\big)(1-\delta)^2}, \tag{5}$$

where R denotes any of R_{E}, R_Ω, $\sqrt{P_{\max}}R_{\mathcal{A}}$ (all equal here, with $P_{\max} = P$). Since $\beta(k,\varepsilon,\delta,n) \to \frac{1}{2+k(n-1)} < \frac{1}{2n}$ as $(\varepsilon,\delta) \to (0,0)$ whenever $k > 2$, every $k > 2$ admits explicit (ε,δ) for which all three *inequalities in* (3) *fail.*

Proof. Divide the bound of Theorem 5.3 by $qR^2/P = qh^2(1-\delta)^2/P$ and substitute a, b:

$$\frac{P\|u\|_{L^\infty}}{qR^2} \le \frac{P\big(ah^2(1-\delta^2) + b(\varepsilon h)^2\big)}{qh^2(1-\delta)^2} = \frac{(1-\delta^2)+\frac{k}{2}\varepsilon^2}{(2+k(n-1))(1-\delta)^2}.$$

For $k > 2$, $\frac{1}{2+k(n-1)} < \frac{1}{2n} \iff 2n < 2 + k(n-1) \iff k > 2$; continuity of β finishes. By Lemma 5.4 the three statements (3) coincide on this family, so all fail together. □

Corollary 5.6 (Exact rational instance)**.** *For $n = 3$, $k = 4$, $\delta = \frac{3}{20}$, $\varepsilon = \frac{1}{5}$:*

$$\frac{P_{\max}\|u\|_{L^\infty}}{qR^2} \ \le\ \frac{1-\frac{9}{400}+\frac{2}{25}}{10\cdot\left(\frac{17}{20}\right)^2} \ =\ \frac{423/400}{2890/400} \ =\ \frac{423}{2890} \ <\ \frac{1}{6}, \qquad \frac{1}{6} - \frac{423}{2890} = \frac{88}{4335}.$$

A bounded Lipschitz domain, constant scalar coefficient, flat Dirichlet mouth: every hypothesis of the conjecture holds and the coefficient is 12.2% below the conjectured sharp constant.

Corollary 5.7 (The infimum is zero). *$\inf\Big\{P_{\max}\|u\|_{L^\infty}/(qR^2)\Big\} = 0$, the infimum over all admissible $(\Omega, \Sigma, \mathcal{A}, q)$, in each of the three metrics; it is not attained.*

Proof. Take $\varepsilon_k^2 = 1/k$ and $\delta_k = 1/k$ in (5): $\beta = \big(\frac{3}{2} - k^{-2}\big)/\big((2 + k(n-1))(1 - k^{-1})^2\big) \to 0$ as $k \to \infty$. The ratio is strictly positive for every fixed datum by Lemma 1.2. □

Proposition 5.8 (Cusp sharpening: the untruncated horn). *Let $k > 2$ and let $\Omega_{k,\varepsilon}$ be the full cusp horn ($\delta = 0$), with Σ the mouth. Define V_0 as the H^1-closure of Lipschitz functions on $\overline{\Omega}$ vanishing on a neighborhood of $\overline{\Sigma}$, and let $u \in V_0$ solve (1) for $\varphi \in V_0$. Then u exists and is unique,*

$$\|u\|_{L^\infty} \;\le\; ah^2\Big(1 + \tfrac{k}{2}\varepsilon^2\Big), \qquad \frac{P_{\max}\|u\|_{L^\infty}}{qR^2} \;\le\; \frac{1 + \frac{k}{2}\varepsilon^2}{2 + k(n-1)},$$

with $R = h$, and this is $< \frac{1}{2n}$ exactly when $\varepsilon^2 < \dfrac{(n-1)(k-2)}{nk}$. (For $n = 3$, $k = 4$: $\varepsilon^2 < \frac{1}{3}$.)

Proof. Well-posedness. For Lipschitz φ vanishing near $\overline{\Sigma}$, the segment from (x, y) to (h, y) lies in $\overline{\Omega}$ (g nondecreasing), so $\varphi(x, y) = -\int_x^h \partial_t\varphi(t, y)\,dt$ and, by Cauchy–Schwarz and Fubini, $\int_\Omega \varphi^2 \le h^2 \int_\Omega |\partial_x\varphi|^2$. The Poincaré inequality passes to the closure V_0; Lax–Milgram applies, and Lemma 1.2 holds verbatim (the truncations used there preserve the approximating class).

Variational identity for v. Fix Lipschitz φ vanishing near $\overline{\Sigma}$ and $s \in (0, h)$. On the Lipschitz domain $\Omega \cap \{x > s\}$, Gauss–Green for $\varphi P\nabla v$ gives $\int_{\{x>s\}} \nabla\varphi \cdot P\nabla v = \int_{\{x>s\}} q\varphi + \int_\partial P(\partial_\nu v)\varphi$; the wall term vanishes by Proposition 5.2(ii), the Σ term because $\varphi \equiv 0$ there, and the face $\{x = s\}$ contributes at most $P \sup|\nabla v|\, \|\varphi\|_\infty\, \kappa_{n-1} g(s)^{n-1} \to 0$ as $s \downarrow 0$. Dominated convergence gives $B[v, \varphi] = \int q\varphi$, and by density the identity holds for all $\varphi \in V_0$; hence $B[u - v, \varphi] = 0$ on V_0.

Admissible test function. Let $M = b(\varepsilon h)^2$ and $\psi = v + M$. On $\overline{\Omega}$ one has $\rho \le g(x) \le \varepsilon h$, so $\psi \ge a(h^2 - x^2) \ge 0$ everywhere. If u_j are Lipschitz approximants of u vanishing near $\overline{\Sigma}$, then $(u_j - \psi)^+$ vanishes near $\overline{\Sigma}$ as well (there $u_j = 0$ and $\psi \ge 0$) and is Lipschitz; by Stampacchia truncation continuity, $(u - \psi)^+ \in V_0$.

Conclusion. Testing $B[u - v, \cdot] = 0$ with $(u - v - M)^+ = (u - \psi)^+$ gives, as in Theorem 5.3, $u \le v + M \le ah^2 + b(\varepsilon h)^2$ together with $u \ge 0$. Lemma 5.4 holds verbatim with $\delta = 0$, so $R = h$. Finally

$$\frac{1 + \frac{k}{2}\varepsilon^2}{2 + k(n-1)} < \frac{1}{2n} \iff 2n + nk\varepsilon^2 < 2 + k(n-1) \iff \varepsilon^2 < \frac{(n-1)(k-2)}{nk}.$$

□

Remark 5.9 (Status of the cusp realization). Proposition 5.8 is logically auxiliary: every closure statement above — Theorem 5.5, Corollaries 5.6 and 5.7 — rests entirely on the truncated Lipschitz family. The cusp version uses the nonstandard boundary realization V_0 at the tip and is flagged for independent functional-analytic review.

6 The dichotomy within the horn family: the cone as exact transition

Proposition 6.1 (Star deficit of the horn wall). *On the wall of $\Omega_{k,\varepsilon,\delta}$ (or of the untruncated horn), for $x_0 = (x_*, 0)$ with $\delta h \le x_* < h$, the star integrand satisfies*

$$(z - x_0) \cdot \nu \;\propto\; g(x) - (x - x_*)\, g'(x) \;=\; \Big(1 - \tfrac{k}{2}\Big) g(x) \;+\; x_*\, g'(x),$$

and on the end face F_δ (normal $-e_1$), $(z - x_0) \cdot \nu = x_* - \delta h \ge 0$. Hence: $k \le 2$ implies (4) at every axis point x_0, while for every $k > 2$ the wall integrand is negative for all $x > x_*\,k/(k-2)$, so (4) fails at every interior x_0 once the horn is long enough.

Proof. With $z = (x, g(x)\hat{y})$ and outward normal $\propto (-g'(x), \hat{y})$, $(z - x_0) \cdot \nu \propto -(x - x_*)g'(x) + g(x)$; substituting $xg' = \frac{k}{2}g$ gives the display. For $k \le 2$ both terms are nonnegative ($g, g' \ge 0$). For $k > 2$, $(1 - \frac{k}{2})g + x_* g' = g\big(1 - \frac{k}{2} + \frac{kx_*}{2x}\big) < 0$ iff $x > x_* k/(k-2)$. □

Theorem 6.2 (Dichotomy and pinch at the cone). *Fix $n \geq 2$ and the horn family of Definition 5.1 with $P = P_{\max}$.*

(i) ***Sub-conical ($k \leq 2$): the distance bound holds with the conjectured constant.*** *For every ε, δ,*

$$\|u\|_{L^\infty} \;\geq\; \frac{q\,R^2}{2n\,P_{\max}}, \qquad R = h(1-\delta).$$

(ii) ***Super-conical ($k > 2$): it fails.*** *For (ε, δ) small (explicitly, whenever $\beta(k, \varepsilon, \delta, n) < \frac{1}{2n}$, e.g. the window of Proposition 5.8), the coefficient is strictly below $\frac{1}{2n}$.*

(iii) ***Cone ($k = 2$): pinched at the conjectured constant.***

$$\frac{1}{2n} \;\leq\; \frac{P_{\max}\|u\|_{L^\infty}}{qR^2} \;\leq\; \frac{1-\delta^2+\varepsilon^2}{2n\,(1-\delta)^2} \;\xrightarrow[(\varepsilon,\delta)\to 0]{}\; \frac{1}{2n}.$$

Within the power-law horn family, $k = 2$ is thus the exact transition: the star condition — a sharp sufficient *hypothesis, by Theorem 4.2 and Proposition 4.3 — controls the sub-conical regime at the constant $1/(2n)$ in every dimension, and beyond the cone its removal is accompanied by arbitrarily severe failure of the distance bound (Corollary 5.7). Necessity of the star condition across all geometries is not asserted.*

Proof. (i) By Proposition 6.1, every axis point $x_0 = (x_*, 0)$, $x_* \in (\delta h, h)$, satisfies (4); Theorem 4.2 with $\mathcal{A} = P_{\max}I$ and $d_{\mathrm{E}}(x_0, \Sigma) = h - x_*$ (Lemma 5.4) gives $u(x_0) \geq q(h-x_*)^2/(2nP_{\max})$. Since u is continuous in Ω, $\|u\|_{L^\infty} \geq \sup_{x_* \downarrow \delta h} q(h-x_*)^2/(2nP_{\max}) = qR^2/(2nP_{\max})$. (ii) is Theorem 5.5 / Proposition 5.8. (iii) The lower bound is (i) at $k = 2$; the upper bound is (5) at $k = 2$, where $2 + k(n-1) = 2n$. ☐

7 Mechanism and internal consistency

Remark 7.1 (Two exact one-dimensional balances; the graded-resistance identity). Theorem 2.1 is the conductivity-graded balance: sink q per unit *length*, conductivity $A(x)$ varying; its sharp constant is $1/2 = 1/(2n)|_{n=1}$. The horns realize the volume-graded balance: sink q per unit *volume* in a tube of cross-sectional volume $\mathfrak{A}(s) = \kappa_{n-1} g(s)^{n-1}$. The far-end value of the exact solution reproduces the series (graded-tube) functional identically: with $V(s) = \int_0^s \mathfrak{A}$,

$$\frac{q}{P}\int_0^h \frac{V(s)}{\mathfrak{A}(s)}\,ds = \frac{q}{P}\int_0^h \frac{s}{\frac{k(n-1)}{2}+1}\,ds = \frac{q\,h^2}{P\,\big(2+k(n-1)\big)} = v(0,0).$$

The conjectured functional $q\,d(\cdot,\Sigma)^2/(2nP_{\max})$ extrapolated the first balance; it retains the distance but discards the volume grading $V/\mathfrak{A}$, and the horns are exactly the geometries on which the two functionals separate without bound.

Remark 7.2 (Theorem 3.3 on the horns). Every ball inside a horn lies in the slab $\{|y| < \varepsilon h\}$, so its radius is at most εh and Theorem 3.3 asserts only $\|u\|_{L^\infty} \geq q\varepsilon^2 h^2/(n(n+2)P_{\max})$ — far below the actual value $\approx ah^2$ for small ε, and consistent with Theorem 5.3. The mean bound and the falsification never touch.

Remark 7.3 (Corollary 4.4 on the horns). On a horn, $\operatorname{dist}(z, \partial\Omega) \leq g(x) \leq \varepsilon h$ for every z, so the unconditional bound of Corollary 4.4 makes assertions only at the scale εh, again consistent with Theorem 5.3. The closure of the distance conjecture requires the wall to *reflect*: if the wall is made Dirichlet it joins Σ, all distances to the supply collapse to $O(\varepsilon h)$, and no conflict can arise.

Remark 7.4 (Verification). Every displayed algebraic identity and rational inequality above — the 1D flux law and its bounds (Theorem 2.1), the ball-mean constant $2r^2/(n+2)$ and the resulting $n(n+2)$ (Theorem 3.3; 15 at $n = 3$), the identities of Proposition 4.1, the horn solution and wall-flux identities (Proposition 5.2), the coefficient formulas and windows, the instance $423/2890 < 1/6$, the star deficit (Proposition 6.1), and the cone marginality $1/(2+k(n-1))|_{k=2} = 1/(2n)$ — has been machine-verified in exact symbolic/rational arithmetic (28 independent checks; companion artifact `p2-softcheck.py`, exit 0). Finite-volume corroboration of the horn coefficients exists in the companion harness and is deliberately excluded here: this paper contains forced material only.